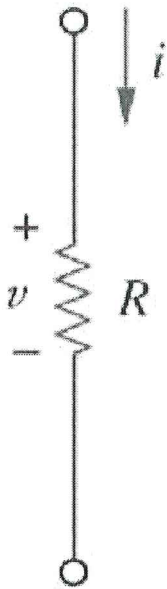


L#03: Ohm's law, node, branch, loop

2018-02-06

Ohm's law

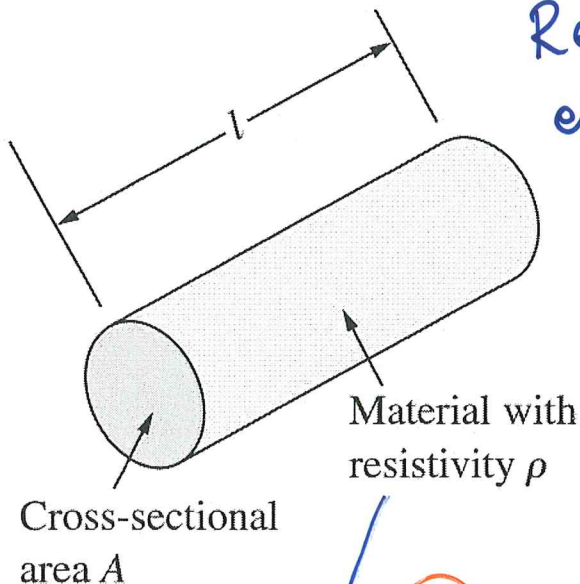
Law, sign convention, and power



Ohm's law states that
voltage (v) = k current (i)
 Δ
constant

$$v = k i \Rightarrow \boxed{v = R \cdot i}$$

Δ
 $\downarrow R$

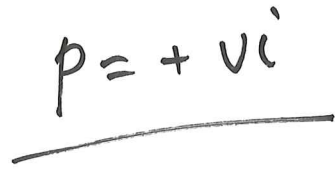


Resistor is a circuit element that resist current.

R : resistance, Ohm (Ω)

$$R = \rho \cdot \frac{l}{A}$$

$$\Omega \cdot \cancel{m} \cdot \frac{\cancel{m}}{\cancel{m^2}}$$



①

②

③

→ π^0

$P \geq 0$, Resistors consume energy.

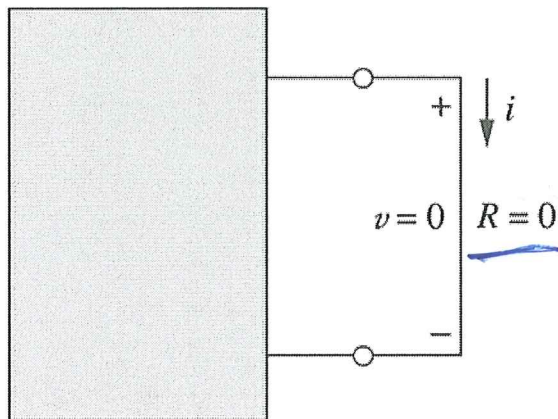
Resistivity ρ & Resistance R

Material	Resistivity ($\Omega \cdot m$)	Usage
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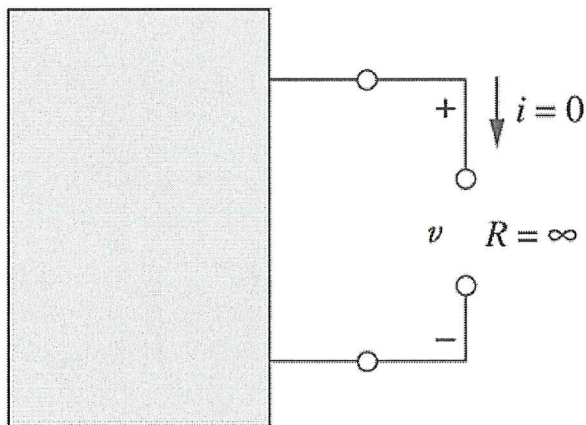
Copper	1.72×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semi-conductor
Silicon	6.4×10^2	Semi-conductor
Paper	1×10^{10}	Insulator
Teflon	3×10^{12}	Insulator

$$R = \rho \frac{l}{A} = \left[\Omega \cdot m \frac{m}{m^2} \right] = [\Omega]$$

Resistance R – extremes



$R \rightarrow 0, i \rightarrow \infty$
short circuit



$R \rightarrow \infty$
open circuit

Conductance G

reverse of resistance

$$G = \frac{1}{R}, \text{ Siemens } [S].$$

Node, branch, loop

Branch:

each two terminal circuit elements is a branch.

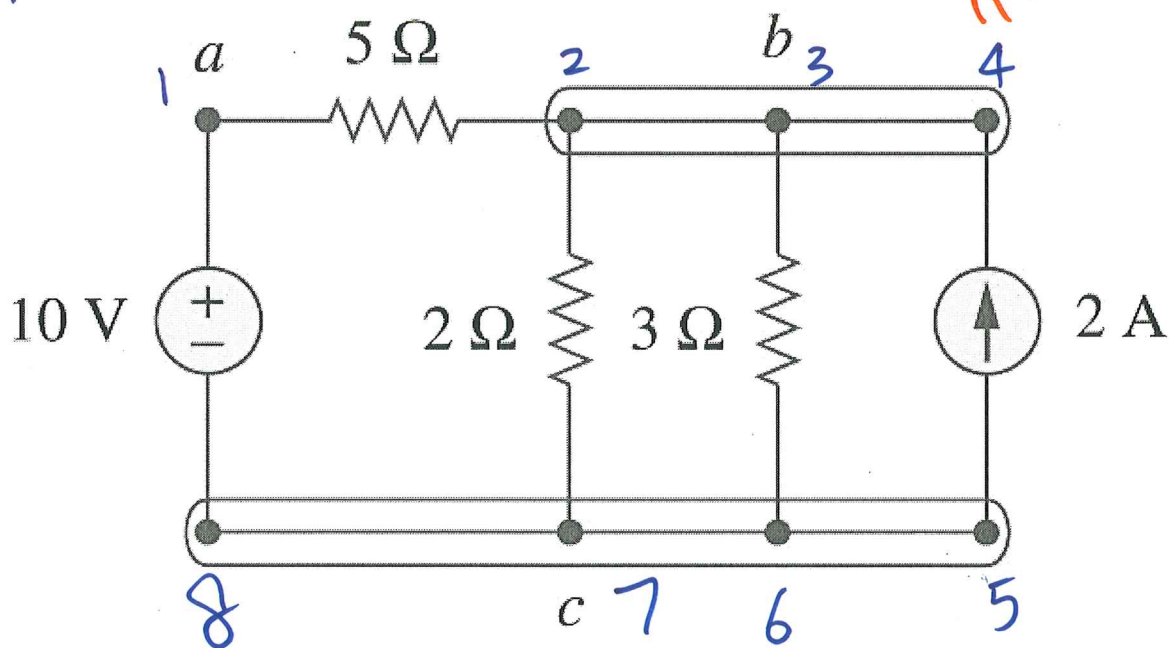
Node:

point of connection of 2 or more elements. Same voltage

Loop:

closed path

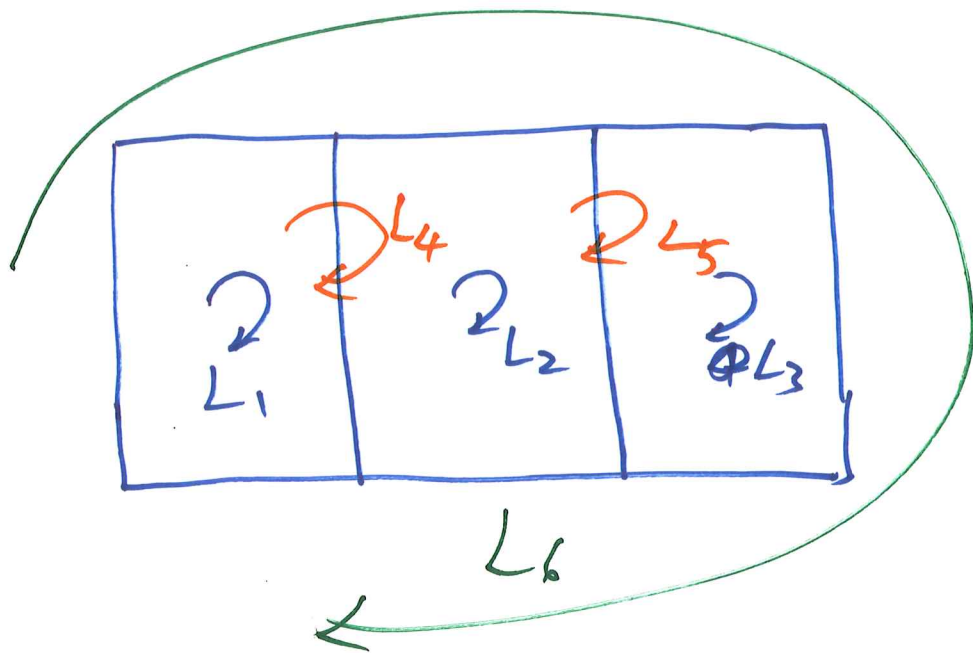
mesh: loops not contain any other loops.



$$L_1(1, 2, 7, 8), L_2(2, 3, 6, 7)$$

$$L_3(3, 4, 5, 6), L_4(1, 3, 6, 8)$$

$$L_5(2, 4, 5, 7), L_6(1, 4, 5, 8)$$



$$L_{pS_1} \{ L_1 \} \quad L_{pS_2} \{ L_1, L_2 \}.$$

① ②

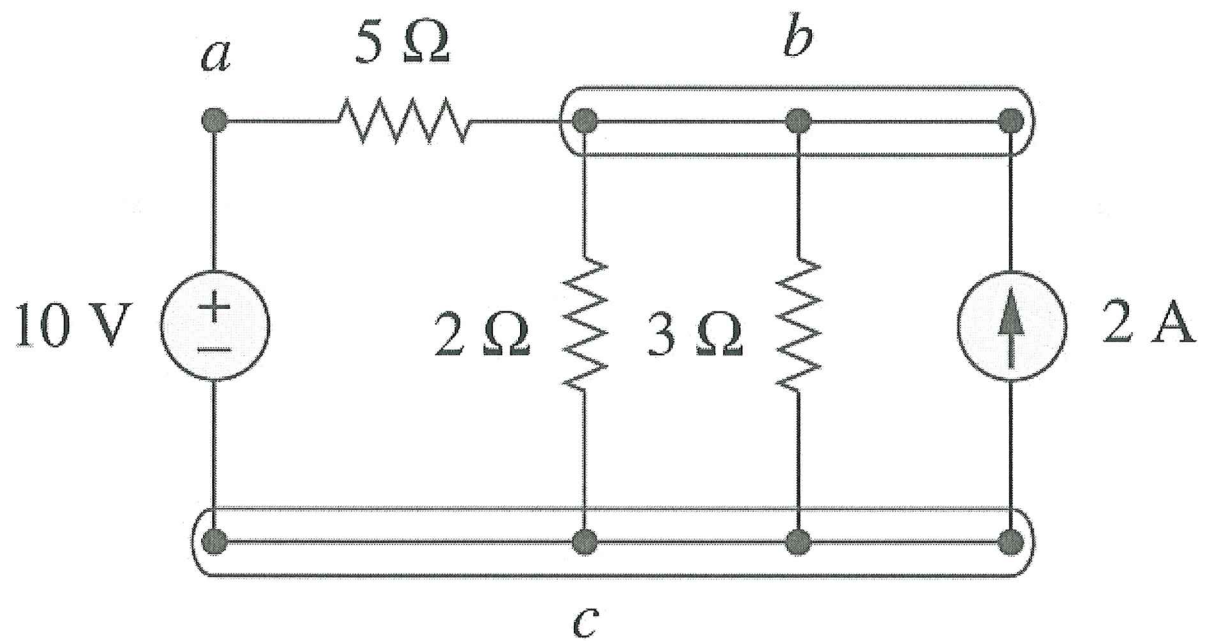
$$L_{pS_3} : \{ L_1, L_2, L_3 \}. \quad ③$$

$$L_{pS_4} : \{ L_1, L_2, L_4 \}.$$

ind. Loop set = $\frac{\text{number of meshes.}}{1}$

max number of loops in

Independent loop sets



$$b = l + n - 1$$

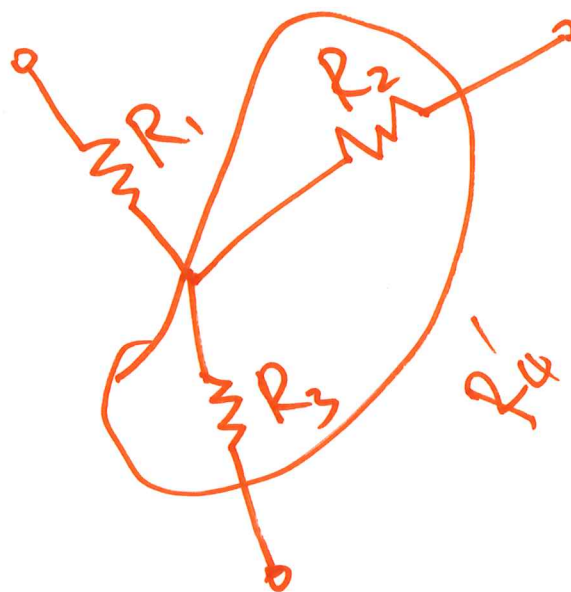
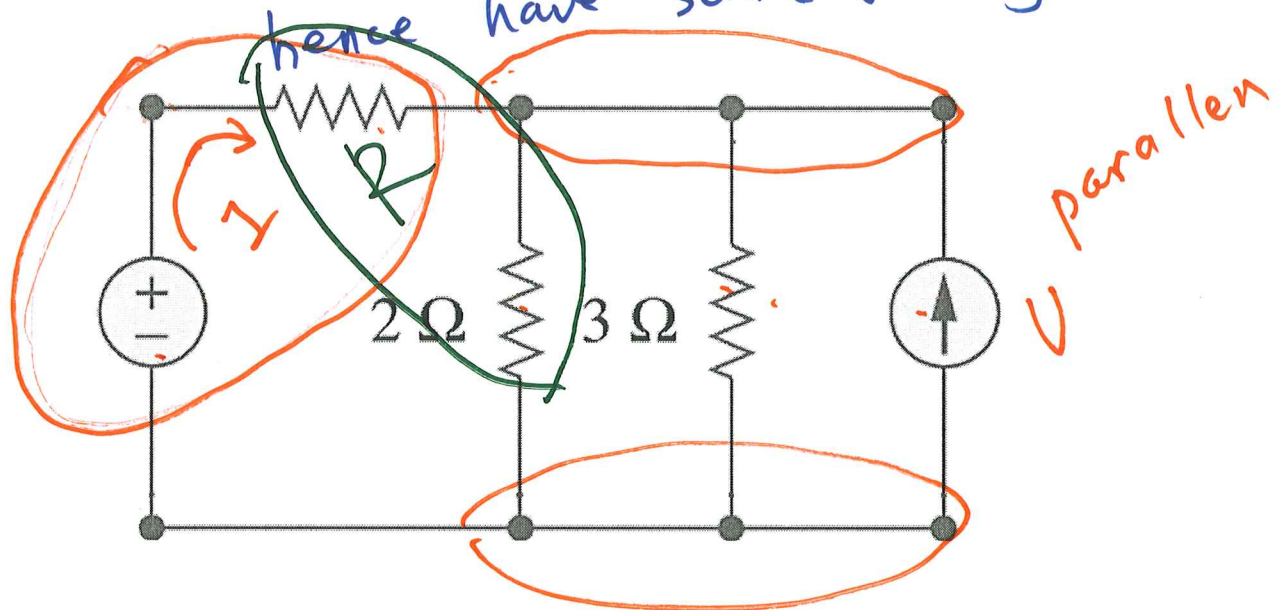
Handwritten values for the equation above:

- $b = 5$ (with an arrow pointing to b)
- $l = 3$
- $n = 3$

Series and Parallel

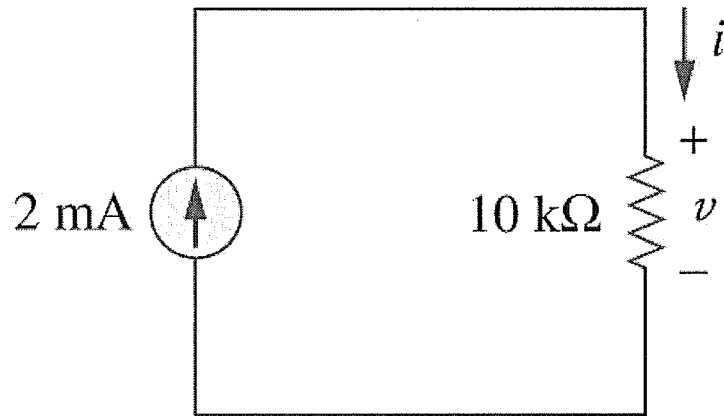
Series: 2 or more elements share one single exclusive node, hence have same current.

Parallel: 2 or more elements share two nodes hence have same voltage.



Example #1:

For the circuit shown, calculate the voltage v , conductance G and power p .



Example #2:

Find the number of nodes, branches and independent loops.

